```
$text = 'I contain no digits.';
print 'Filter: ' . preg_filter('/\d+/', '!', $text);
print PHP_EOL;
print 'Replace: ' . preg_replace('/\d+/', '!', $text);
```

```
Output:
Filter:
Replace: I contain no digits.
```

As an alternative to the simple string replacement in `preg_filter()` and `preg_replace()`, it is possible to use a callback function and `preg_replace_callback()`. Using this function, each time a match is made, the callback is invoked with an array argument that contains the matched portion of the sample string (and any capture groups if specified); the callback should return the desired replacement string. `preg_replace_callback()` is otherwise very similar to `preg_replace()`, and its other parameters are identical – it will accept strings or arrays of strings in the `$pattern` and `$text` positions and has support for offset, count, limit, and flag parameters. Here's a quick demonstration:

```
$text = 'The cat chased the bird, the dog chased the cat.';
$pattern = '/dog|cat|bird/';
print preg_replace_callback($pattern, function (array $matches): string {
    print 'callback invoked' . PHP_EOL;
    return match ($matches[0]) {
        'dog' => 'black dog',
        'cat' => 'white cat',
        'bird' => 'blue bird',
    };
}, $text);
```

```
Output:
callback invoked
callback invoked
callback invoked
callback invoked
The white cat chased the blue bird, the black dog chased the white cat.
```